

LETEK DCG / Bachoco - chicken manure

DEHYDRATION SYSTEM MASS-ENERGY BALANCE

3.22 TPH wet chicken manure digestate cake (screw-pressed)
 4.33 C.Y./hr @ 1485 lb/C.Y. and 70% moisture
 8x40 Single-pass rotary drum · XD-96 (8'x40')
 Elevation 30 feet (above sea level)

Fuel to Burner	
Propane	
19,900	Btu/lb
366	lb/hr
7.28	MMBtu/hr

Make-Up Air	
5,704	SCFM
85	deg F

Mixed Inlet Gases	
7,445	SCFM
900	deg F

1,589 Btu's/lb evaporation (rigorous)



Exhaust Gases	
12,934	ACFM
50	hp fan

Dryer Outlet Gases	
12,934	ACFM
230	deg F

Combustion Air	
1,675	ACFM
85	deg F
30%	X-S air

Wet Feed	
70.0%	moisture
1,929	lb/hr solids
4,501	lb/hr water
6,430	lb/hr total

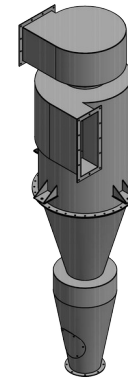


EVAPORATION RATE
 4,161 LB/HR H₂O

XD-96 (8'x40') · 40 ft long · 8.7 MMBtu/hr burner

Dry Fuel (product burned)	
0	lb/hr total

Net Dry Product	
2,269	lb/hr total



Dry Material	
15.0%	moisture
1,929	lb/hr solids
340	lb/hr water
2,269	lb/hr total
180	deg F

